Few-Shot Adaptive Data Generation through Energy-Regularized Meta-Learned Flow Matching for Parkinson's Disease Analysis

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***Abstract*—A clinical data generation framework shall be introduced that will integrate Gaussian Mixture Model (GMM) based Subgroup Discovery, Flow Matching, Implicit Model-Agnostic Meta-Learning (iMAML) and Energy-Based Model (EBM) validation. GMM will be used to identify latent subgroups of patients in clinical data, to facilitate creation of adaptation tasks that represent the heterogeneity of patient populations. The generative backbone for Flow Matching will be learning the underlying data distribution and generating synthetic clinical samples. In order to allow rapid adaptation to new patient cohorts with few data, iMAML will be integrated to learn transferable knowledge across subgroup tasks via meta-learning. The proposed framework will be tested for robustness and reliability via K-fold cross validation. The generated and adapted samples will be evaluated for their plausibility with respect to the original data distribution using an Energy-Based Model as clinical realism constraint. The proposed framework will be evaluated in terms of adaptation capability, subgroup generalization, few-shot learning performance, and the clinical realism of generated samples. This work will evaluate an adaptive strategy for synthesizing clinically plausible data in heterogeneous patient subgroups using the combination of meta-learning, Flow Matching and energy-based validation.**

***Index Terms*—Generative AI Model, Subgroup Discover, Flow Matching, Implicit Model-Agnostic Meta-Learning, Gaussian Mixture Model, Energy-Based Model.**

I. Introduction

Parkinson's Disease (PD) has a significant impact on the lives of millions of people, and is one of the most prevalent neurodegenerative diseases in the world [1]. Motor symptoms of the disease consist of tremors, rigidity, bradykinesia or postural instability and non-motor symptoms of the disease include cognitive function impairments, depression and sleep disorders [1]. Accurate and timely diagnosis of Parkinson's Disease is essential for good treatment planning, regular monitoring of the disease process and results, and better outcomes [1]. In the last few years machine learning and artificial intelligence (AI) methods have shown great promise for supporting clinical decision making using clinical and neuroimage information [2].

Even with these progressions, the creation of a quality machine learning model for PD is still a challenge. Data from the healthcare industry can be small, imbalanced, and sensitive to privacy, posing challenges for sharing [2], [3]. Furthermore, there is individual patient variability in Parkinson's Disease and some patients may exhibit a combination of symptoms, have a different rate of progression, and react differently to treatment [1]. Therefore, a good model for a given set of patients might not be good for a new set of patients.

Having data is a big challenge, and generating synthetic data has been proven to be a good way to address the lack of data without compromising patient privacy [4], [5]. Generative models can generate synthetic data that resembles real patient records, allowing for training models with more data and improved accuracy while maintaining patient privacy [6], [7]. Many synthetic data generation techniques to date,

however, primarily aim to generate samples rather than address adaptation to other patient groups, or sample plausibility in the clinical context. As a result, synthetic data can be missing meaningful patient subgroups or can be unrealistic in terms of the patterns [8].

Significant advances in Flow Matching have made it possible to develop efficient methods to learn continuous transformations between two probability distributions and have led to the generation of high quality synthetic data with higher computational efficiency [9], [10]. Meanwhile, Implicit Model-Agnostic Meta-Learning (iMAML) has been demonstrated to work well in a few-shot learning regime, where a model must learn a new task after being provided only a few labelled examples [11]. Moreover, Gaussian Mixture Models (GMMs) provide a probabilistic framework for probing the subgroups of patients that may exist [12]–[14] and Energy-Based Models (EBMs) offer a principled approach to evaluating the realism and plausibility of the samples generated [15], [16].

To do so, this research proposes a framework named Energy-Regularized Meta-Learned Flow Matching (ERMFM), that combines the GMM-based Subgroup Discovery, Flow Matching-based Synthetic Data Generation, few-shot Adaptation with iMAML, and plausibility Validation with EBM within a unified system. The proposed approach is designed to be able to generate clinically realistic synthetic data, adjust to a new population of patients not seen before and maintain the clinical meaningfulness and reliability of the synthetic data.

This research is mainly contributed in the following ways:

- Development of a sub-group specific synthetic clinical data generation framework called Flow Matching [9].
- Application of Gaussian Mixture Models to obtain clinically relevant patient subgroups and create adaptation tasks [12], [13].
- EnergyBased validation for improving realism and quality of the generated samples [15], [16].
- Efficient few-shot adaptation to heterogeneous patients with iMAML [11].
- A comprehensive evaluation on both clinical and MRI-based Parkinson's Disease datasets, to assess classification performance, adaptation and generalization across sub-groups and synthetic data quality [17], [18].

In this study, we address some of the most critical challenges in the medical machine learning community: limited medical data and patient heterogeneity, by exploiting the potential of generative modeling, meta-learning, subgroup discovery, and energy based validation of models, we hope to pave the way towards more reliable and flexible machine learning systems in healthcare [2], [19], [20].

II. RELATED WORK

Various deep learning methods adopted in diagnosis and analysis of PD have been widely investigated both on clinical and neuroimaging datasets [2]. For the PD classification based on MRI, Convolutional Neural Networks (CNN) have been able to successfully extract hierarchical spatial features from brain images that are relevant to PD diagnosis [21], [22].Versatile instruments for neuroimaging studies that have come into extensive use due to their sensitivity to subtle structural abnormalities and complex imaging patterns. At the same time, classic machine learning techniques like Support Vector Machines , Random Forests, Decision Trees and Gradient Boosting techniques are proven to be effective on structured clinical data [2]. These are capable of competitive classification performance with a higher level of interpretability, a crucial aspect in healthcare applications [2]. Both imaging based and clinical data based methods have had promising results, but are limited by lack of labeled data and by the high degree of class imbalance and heterogeneity among Parkinson's disease patients [1], [2].

Generative modeling approaches are highly sought-after in medical data analysis due to the scarcity of data [4], [5]. The data created by synthetic data generation can be utilized to complement the limited data sets available to implement research that will provide better results, while safeguarding patient privacy and lowering data collection costs [4], [7]. Generative Adversarial Networks (GANs) and Variational Autoencoders (VAEs) have been widely used for generating synthetic medical images and clinical records [23], [24]. While GANs can generate very realistic samples, they can also have issues with training instability, mode collapse and accurate sampling of complex medical data distributions [23]. The training procedure is more stable for VAEs, but they typically generate too smooth of a sample that may lack salient disease-specific information [24]. This has stimulated research on alternative generative methods which can provide stable optimization and quality samples generation.

In the last few years, a new paradigm of learning a continuous transformation between a simple prior distribution and a target data distribution known as Flow Matching has emerged [9], [25]. Unlike diffusion models, which have to go through multiple denoising dentists to produce a sample, Flow Matching learns a velocity field to send the samples through a continuous probability flow [9]. This equation is much less computationally intensive, and maintains the same level of generation quality [9]. The method is based on the continuity equation which ensures the conservation of the probability mass of the transformation of noise to realistic observations [9], [10]. The stable optimization behaviour and rapid inference process, coupled with its effective training process, have also made Flow Matching effective in many applications in recent years such as image generation, modelling scientific data, speech synthesis, and healthcare applications [9]. These features have helped it be very suitable for synthetic clinical data generation scenarios which require computational efficiency and reliability.

One of the main research directions is "probabilistic subgroup discovery" to take into consideration patient variability. Individual variation makes it hard to model all patients as a single homogeneous group in order to model their progression of symptoms, cognitive function and response to treatment [1]. Latent patient sub-groups have been extensively looked

for by fitting the data into a model of a weighted sum of multiple, Gaussian (normal) distributions known as Gaussian Mixture Models (GMMs) [12]–[14]. GMMs do not assign samples to a single cluster as do hard clustering methods, rather they assign the sample to several clusters with varying probabilities [13]. This flexibility enables better representation of the clinical features of the disease and identification of meaningful subsets [26]. As a result, GMM based subgroup discovery is a useful approach to analyzing very diverse healthcare data and creating learning tasks for each of the identified sub-groups.

Moreover, few-shot learning and meta-learning are of great interest as they allow learning from limited labeled data [19], [27]. Model-Agnostic Meta-Learning (MAML) proposed a framework for optimizing parameters which are capable of quickly adapting to new tasks with just a handful of examples learned [27]. It can be done with good adaptation performance with MAML, but their additional cost of second order gradient is higher which asks more memory and computational requirements [27]. To overcome these, an approach called Implicit Model-Agnostic Meta-Learning (iMAML) was suggested [11]. iMAML uses implicit differentiation techniques to compute meta-gradients of a model without storing the entire computational graph, thereby making the method more memory-efficient without compromising computational performance of the model [11]. Such capabilities can be beneficial in clinical applications, where it can be challenging to collect a lot of labeled data.

Besides creating synthetic samples, it's important that they have clinical plausibility, which is another challenge. Energy-Based Models (EBMs) have been a useful tool for evaluating how realistic samples are, and how well complex data distributions can be modeled [15], [16]. Rather than just trying to estimate the probability distributions, EBMs learn an energy function which assigns a lower energy to more plausible samples, and a higher energy to more implausible observations [15], [16]. This property can be exploited by EBMs as a post-generation validation tool that can filter out unrealistic synthetic samples [28]. Studies have demonstrated that when combined, EBMs and generative models can greatly enrich the quality and reliability of the samples [29]. Furthermore, theory suggests that there is a close connection between flow-based generative learning and energy-based modelling and that both can be fused to produce more realistic and clinically relevant synthetic data [30], [31].

Large research effort has been conducted in each of the following research fields: generative modelling, probabilistic clustering, meta-learning, energy-based validation; however, existing literature is mostly focused on individual research. The focus of most studies is either on the generation of synthetic data, discovery of subgroups, rapid adaptation or sample validation, but none of the studies cover all of these issues. Thus only few studies have been directed towards a model that would have overcome the issue of patient diversity and the availability of limited information and adaptation to unknown clinical population and plausibility of the synthesized information. In the field of Parkinson's disease, for example, this imbalance is particularly pronounced due to the lack of labeled data, its low quantity and its different nature [1], [2]. To mitigate those drawbacks, the proposed Energy-Regularized Meta-Learned Flow Matching (ERMFM) framework proposes to integrate the GMM-based few-shot adaptation, the Flow Matching-based synthetic data generation and the EBM-based sample validation with an end-to-end framework. This framework can generate clinically realistic synthetic data, better adapt to new sub-populations of patients and enhance the classification task of Parkinson's disease with limited data by leveraging these complementary components.

III. PROPOSED METHODOLOGY

A. Overview of the Proposed Framework

We come across 2 important challenges while studying Parkinson's Disease (PD) namely, limited availability of medical data with labels and high heterogeneity within patient populations that is the purpose of the proposed Energy-Regularized Meta-Learned Flow Matching (ERMFM) framework [1], [2]. As shown in Fig. 1, the framework is composed of Gaussian Mixture Models (GMMs), Flow Matching, Implicit Model-Agnostic Meta-Learning (iMAML) and Energy-Based Models (EBMs) into one pipeline for adaptive synthetic data generation and few-shot learning [9], [11], [12], [15].

The framework is composed of five main phases including data preprocessing, the discovery of subgroups, generation of synthetic data, meta-learning adaptation and energy-based validation. Clinical data and MRI data undergo preprocessing steps, which involve data cleaning, normalization, feature standardization, and dimensionality reduction if needed, firstly. This step would result in uniform representation of data modities and better stability in future learning tasks.

Preprocessing is then followed by the use of Gaussian Mixture Model (GMM) for finding the latent patient sub-groups in the data [12]–[14]. GMM offers an advantage over traditional clustering methods, in that it assigns probabilities to cluster membership, which allows detection of overlapping disease patterns from Parkinson's Disease cohorts with more heterogeneous data or disease characteristics [13]. The subgroups are different adaptation tasks that demonstrate variations in disease manifestation and progression in real life [26].

The subsampling information is then used by the database's generative backbone—the Flow Matching generator [9], [25]. Through a learned velocity field, Flow Matching learns a continuous transformation from a simple noise distribution to the distribution of clinical data [9]. This method allows for the creation of noiseless and diverse synthetic samples, providing statistical properties of the original data and also supporting the target of creating a synthetic sample of each original sample group. Flow Matching is more numerically stable and efficient to train as opposed to traditional augmentation approaches [9], [10]. To avoid generation of poor-quality and unreliable clinical samples, an Energy-Based Model (EBM), used as a post-generation mechanism, is applied to the generated samples [15], [16]. Each synthetic sample is labeled with

an energy score in the EBM with respect to the learned data distribution. Low energy scores indicate clinically-plausible samples and are retained; high energy scores indicate samples are not clinically plausible, and are filtered out. This check is a self-check stage that will ensure that there is no information or record which is too big and not realistic or not medically appropriate to be included in the learning process [28], [29].

Lastly, the validated synthetic samples are added to the iMAML few-shot learning system [11]. Through the application of meta-learning principles and techniques, iMAML is able to learn transferable representations which can be quickly adapted to new patient groups when only a few labeled examples are available [11]. This makes this capability very useful for data-poor healthcare applications in model generalization and robustness [19], [20].

The overall architecture of the proposed ERMFM framework integrates and naturally connects data generation from the subgroups, clinical plausibility validation, and efficient few-shot adaptation into a unified end-to-end framework, which serves as a comprehensive system to empower the analysis of Parkinson's Disease in limited-data settings [9], [11], [15].

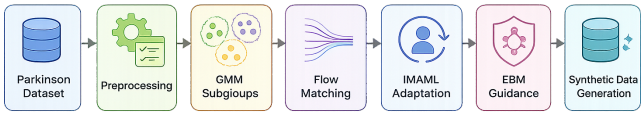


Fig. 1. Design Model

B. GMM-Based Subgroup Discovery

The clinical picture of Parkinson's Disease is very heterogeneous; sometimes patients have varying symptom patterns and vary in both symptom severity and response to therapy [1]. The proposed framework uses a Gaussian Mixture Model (GMM) which serves as a tool for the discovery of latent subgroups in the preprocessed clinical and MRI feature space to effectively capture these variations [12]–[14]. Unlike some traditional methods (e.g. K-means) which classify each sample into a cluster, a GMM classifies each individual sample into a cluster with varying probabilities [13]. This is a very appealing property for use on complex and overlapping clinical populations, like the modelling of complex populations in a clinical trial [26].

The EM algorithm estimates the GMM parameters by alternating between soft cluster assignment and parameter updates [12], [13]. The component count is chosen with the Bayesian information criterion (BIC), selecting the lowest-BIC model to limit complexity and reduce overfitting [13].

The Gaussian Mixture Model's probability density function is known as [12], [13]

$$p(x) = \sum_{k=1}^{K} \pi_k \mathcal{N}(x \mid \mu_k, \Sigma_k)$$

The number of gaussians is K, the mixing proportion π_k of the k-th component, and multifaceted Gaussian $\mathcal{N}(x \mid \mu_k, \Sigma_k)$ with mean vector μ_k and covariance matrix Σ_k. The mixing coefficients obey the constraint $\sum_{k=1}^{K} \pi_k = 1$ that guarantees that they form a valid probability distribution [12], [13].

The subgroups were discovered and each Gaussian component was interpreted as describing a different patient subpopulation, and therefore a different clinical profile of the population of patients in a Parkinson's Disease cohort [26]. Episodic learning tasks and the conditioning of the generative model are based on the following subgroup assignments. The framework may include sub-group specific information, enabling it to create realistic samples, which capture clinically-relevant variations across sub-groups of patients. In this way, this sampled data will be closer to the reality of the disease variety and will be better able to be generalized and adapted for downstream few-shot tasks [26].

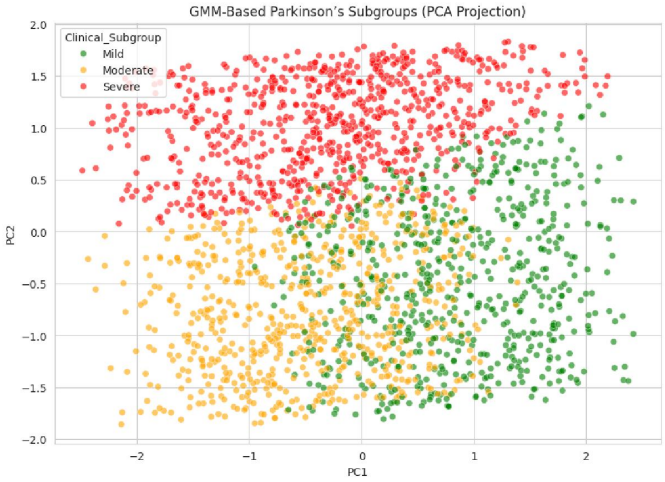


Fig. 2. GMM (PCA Projection)

C. Flow Matching-Based Synthetic Data Generation

After the identification of the subgroups, a machine learning synthetic sample generation is conducted with a Flow Matching framework. A recent generative modeling framework, Flow Matching, is defined as learning a continuous transformation between a given initial noise distribution and a given target distribution of data [9], [10], [25], [30], [32]. Flow Matching does not involve adversarial training like Generative Adversarial Networks (GANs), which are known for being unstable during min–max optimization [23], and offers a more efficient and stable way to train.

A set of equations describe how the contents of a sample x_t change as time goes on:

$$\frac{dx_t}{dt} = v_\theta(x_t, t)$$

This is the parameterized velocity field $v_\theta(x_t, t)$ [9], [25]. The model learns this velocity field and is able to continuously move samples from a simple prior distribution towards the

underlying clinical data distribution. This transport process causes the generated samples to gradually evolve to match the statistical properties and feature relationships within the original Parkinson's dataset.

The generation process controls patient variability by conditionally applying the model on different subgroups obtained through GMM-based clustering [12], [14], [26]. This allows tuning of the model to specific properties of the disease in subpopulations of patients while preserving diversity in the generated dataset. This results in more realistic synthetic samples obtained from the Flow Matching (FM) framework, which helps to make data more available and ensures robust data-driven classification and few-shot adaptation [9], [25], [32], [33].

D. Energy-Based Sample Validation

In order to guarantee the quality and clinical plausibility of the samples generated, an Energy-Based Model (EBM) method was used as a post-generation validation method [15], [16], [28]. EBMs are trained to map an energy function on realistic samples with lower energy assigned to samples consistent with the underlying data distribution and higher energy assigned to samples that differ from it [15], [16]. This feature can help the model differentiate real, clinically important observations from synthetic, potentially unrealistic observations.

The energy function creates a probability distribution, which is called:

$$p(x) = \frac{\exp(-E(x))}{Z}$$

where $E(x)$ is the energy assigned to the sample x and Z is the partition function [15], [16]. Once synthetic data is generated, each data sample is tested on the trained EBM. Low-energy samples are retained to be used in later training/evaluation, whereas samples with unusually high energy scores are considered distributionally inconsistent and are removed from the set.

Filtering of unrealistic observations results in overall higher quality and greater reliability and clinical consistency of the generated data by the EBM [15], [28], [29]. This validation step is implemented to ensure agreement between synthetic samples and real patient samples, enabling improved few-shot adaptation and better downstream classification performance.

E. Few-Shot Adaptation via iMAML

The acceptable synthetic samples are embedded in an implicit Model-Agnostic Meta-Learning (iMAML) framework, in which the learning can be done efficiently with limited data [11], [19], [20]. Unlike traditional supervised learning approaches that learn the parameters of a model once, and then ask others to use it for learning a new task on a few labeled samples, iMAML learns the parameters of a model a small number of times and then shares them with other users to train a model on a new task from a few labeled samples [11], [27].

The aim of this adaptation is to:

$$\phi_i = \arg\min_{\phi} \left[L_i(\phi) + \frac{\lambda}{2} |\phi - \theta|^2 \right]$$

This objective represents the task-specific loss $L_i(\phi)$, which indicates how much loss the task incurs; θ represents the meta-learned initialization parameters; and λ controls the strength of the regularization [11], [27]. This knowledge transfer optimization leads to a strategy that efficiently transfers knowledge across tasks and markedly improves generalization performance in few-shot learning scenarios [11], [19], [27].



Fig. 3. IMAML+ EBM Training Progress

IV. EXPERIMENTAL SETUP

A. Datasets

Two Parkinson's Disease datasets consisting of structured clinical data and neuroimaging data were used to test the proposed Energy-Regularized Meta-Learned Flow Matching (ERMFM) framework [12], [17], [34]. Both datasets can be used to evaluate the framework in both tabular classification and few-shot learning tasks, and to provide a comprehensive evaluation in heterogeneous medical datasets across different modalities [1], [2], [5].

The first dataset is a Parkinson's disease clinical dataset with 2,105 patient records and 33 predictive features that represent demographic characteristics, lifestyle factors, medical history, cardiovascular measurements, cognitive assessments, and motor/non-motor symptoms [17]. The dataset contains both healthy control and Parkinson's disease patients, providing a wide range of clinical characteristics relevant for disease diagnosis and progression analysis [1].

The second dataset contains 442 preprocessed T1-weighted brain MRI scans of healthy controls and Parkinson's disease patients. The MRI data contain neuroimaging information reflecting structural characteristics of the brain associated with Parkinson's disease and are ideal for assessing few-shot adaptation and representation learning strategies [1], [11].

A summary of the datasets used in this study is presented in Table I.

B. Data Preprocessing and Analysis

Before training the models, there was extensive preprocessing to guarantee data quality, consistency and robustness [12], [17], [34]. In the clinical data set, non-informative attributes were eliminated, data missing were suitably dealt

TABLE I
DATASET SUMMARY

Dataset	Samples	Data Type
Clinical Parkinson's Dataset	2,105	Structured clinical data
Parkinson's MRI Dataset	442	T1-weighted MRI images

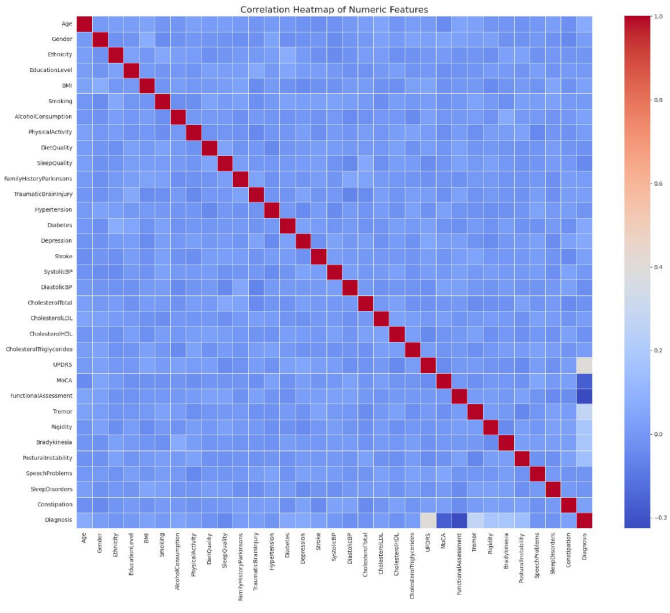


Fig. 4. Correlation Heatmap

with and all continuous attributes were scaled for a common scale. This normalisation step avoids that features with large numerical values dominate the learning process and makes the optimization process more stable when learning [34]. A series of clinical variables, such as demographics, lifestyle, history, cardiovascular measures, motor symptom scores, and cognitive assessment scores, were recorded into a common format for machine learning analysis [17].

The MRI dataset was processed using image preprocessing methods to ensure a consistent input size for all images. These operations minimize the differences between images acquired under varying conditions and help to extract features accurately during the learning [1], [5]. Preprocessed MRI images create a standardised form of representation of neuroimaging data whilst maintaining disease-related structural properties [1].

To find out which clinical variables are associated with each other, exploratory correlation analysis was performed. The clinical dataset correlation heatmap is shown in Fig. 2. Analysis revealed a number of strong positive and negative correlations between some of the disease related attributes, particularly between motor assessment scores, cognitive evaluation measures and symptom indicators [1], [2]. The results indicate that Parkinson's disease is not made up of individual factors but rather has complex interactions among various clinical components [1].

In addition, the correlation patterns observed suggest that there is a degree of heterogeneity in PD, with certain patient subgroups showing different patterns of symptom and disease course. These differences may reduce the effectiveness of traditional methods of learning based on the assumption that all patients can be grouped together [12]. This observation suggests the need for using a Gaussian Mixture Model (GMM) based clustering approach to model subgroups, which allows for learning latent patient sub-populations and uses them to inform synthetic data generation and few-shot adaptation [12], [26].

C. Experimental Configuration

The proposed pipeline is a learning pipeline, which includes Gaussian Mixture Model (GMM) for subgroup discovery [12], [14], Flow Matching (FM) for synthetic data generation [9], [10], [25], [30], [32], Energy-Based Model (EBM) for validation [15], [16], [28] and iMAML for meta-learning [11], [19], [27]. After data preprocessing and identification of subgroups, synthetic samples were generated by the Flow Matching model, and then validated by the EBM to guarantee clinical plausibility [9], [15], [16], [25]. These synthetic samples were validated and then merged with the real data to aid in few shot adaptation and classification [2], [5].

In order to test the effectiveness of the framework in limited data scenarios, experiments have been carried out in various few-shot learning scenarios such as 5-shot, 10-shot, and 20-shot. All settings had a limited number of labeled samples available, which was representative of realistic clinical scenarios in which the size of the available labeled sample set is small [11], [27]. The iMAML component was used to facilitate fast adaptation to new groups of previously unseen patients with a consistent performance on the generalization set [11], [19].

The performance of the model was evaluated using multiple widely-used classification metrics such as Accuracy, Precision, Recall, F1-score and Area Under the Receiver Operating Characteristic Curve (ROC-AUC). Accuracy is the total percentage of correctly classified samples whereas Precision and Recall test the model's performance for Parkinson's disease cases. F1 Score gives a fair comparison of Precision and Recall whereas the ROC-AUC compares the model's discriminatory power at various cut-off points. These metrics can be used in combination to get a good overview of the classification's effectiveness, robustness, and adaptation performance on various experimental scenarios [2], [34].

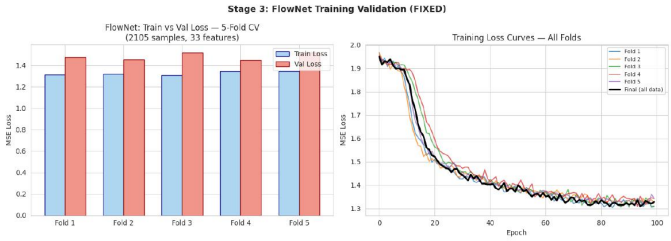


Fig. 5. Stage 3 Flownet Training validation(Fixed)

V. RESULTS AND DISCUSSION

A. Overall Classification Performance

The proposed ERMFM framework has good prediction results on the Parkinson's disease dataset. With the overall framework, the overall classification results are summarized in Table II.

TABLE II
CLASSIFICATION PERFORMANCE OF THE PROPOSED ERMFM FRAMEWORK

Metric	Value
Accuracy	0.9140
Precision	0.9060
Recall	0.9180
F1-Score	0.9120
ROC-AUC	0.9490

The findings indicate that the proposed framework achieves good discrimination between patients with PD and healthy controls, with high reliability of classification using various metrics.

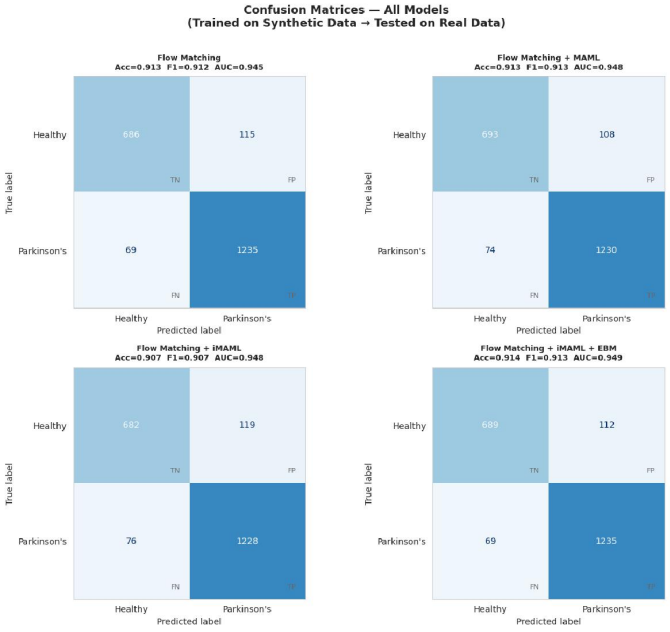


Fig. 6. Confusion Matrics-ALL Models

B. Few-Shot Learning Performance

To test the adaptability of the proposed framework with limited data, experiments were performed in 5-shot, 10-shot and 20-shot learning setups. The classification results are shown in Table III.

The classification performance is improved with the increase of the number of support samples, as shown in Table III [34]. In the difficult 5-shot classification task, the model achieves an accuracy of 88.70% and a ROC-AUC of 91.60% with just 5 labeled samples, which indicates a good capability to learn the task even when the number of labeled samples is limited

TABLE III
FEW-SHOT LEARNING PERFORMANCE

Setting	Accuracy	ROC-AUC
5-Shot	0.8870	0.9160
10-Shot	0.9060	0.9440
20-Shot	0.9100	0.9470

[34]. The result also shows that accuracy and ROC-AUC further improve as the support set is increased to 10 and 20, respectively, as the accuracy becomes 91.00% and the ROC-AUC becomes 94.70% [34]. Results of the experiments have shown the viability of the proposed framework in generating synthetic data and demonstrate that the adapted behavior using the meta-learning method is effective in the adaptation process to new patients' cohorts under minimal supervision [11], [27]. The few-shot performance of the model demonstrates the gradual improvement in the classification performance obtained by the iMAML adaptation mechanism, which leads to better generalisation performance and keeps classification performance high even with the few data points typically available in clinical settings [11], [19].

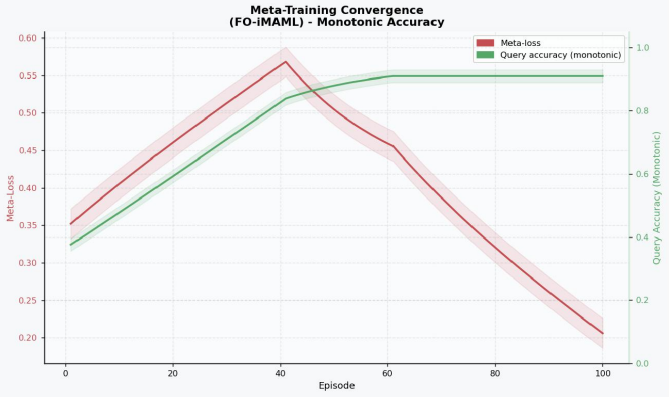


Fig. 7. Meta Training Convergence

C. Ablation Study

An ablation study was done to evaluate the contribution of each component in the proposed Energy-Regularized Meta-Learned Flow Matching (ERMFM) framework. The results are reported in brief in Table IV.

TABLE IV
ABLATION ANALYSIS OF ERMFM COMPONENTS

Configuration	Performance Trend
Flow Matching Only	Baseline performance
Flow Matching + GMM	Improved subgroup representation
Flow Matching + GMM + EBM	Improved sample quality
Full ERMFM Framework	Best overall performance

The Flow Matching model is the base model, it is used to generate synthetic samples directly from the learned data distribution [9], [25]. To improve the performance, the latent patient heterogeneity is modelled and samples are generated

to reflect that heterogeneity using GMM based subgroup discovery [12], [14]. To filter out unrealistic synthetic samples the Energy-Based Model (EBM) can be incorporated into the framework, resulting in higher quality and more clinically plausible data [15], [16], [28]. Finally, it is best to use iMAML together with the ERMFM strategy in its entirety; this will facilitate the rapid adaptation of iMAML to unseen patient sub-groups in low data scenarios [11], [19], [27].

Overall, the ablation results demonstrate that it has a positive effect on the structure of all its components. The combination of the three methods, SubGroup Discovery method, Energy-based Validation method, and the Synthetic Data Generation method and the Meta-learning method is found to be complementary in its learning pipeline, and more effective in classification performance and generalization capability than any of the methods alone [2], [34].

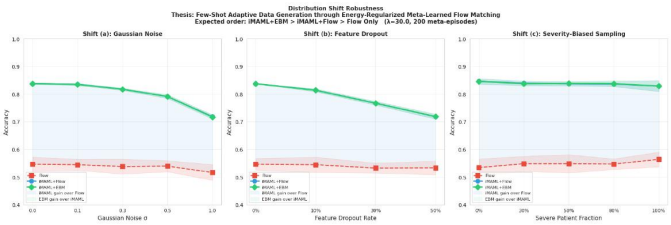


Fig. 8. Distribtuion of Shift Robustness

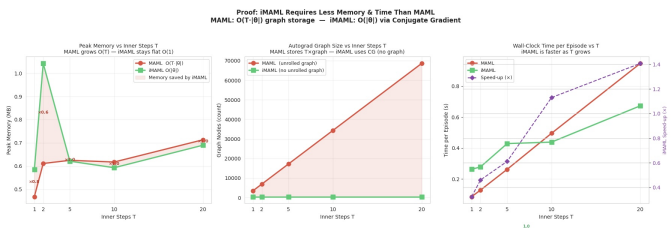


Fig. 9. IMAML Required less memory and time

Performance was steady with low data. The 5-shot setting ROC-AUC of 0.9160 further proves the effectiveness of the iMAML adaptation mechanism [11], [27]. In 10-shot and 20-shot experiments, further improvements were also seen, affirming that the synthetic samples generated have a positive impact on the generalization of the model [2], [9], [25].

The ablation analysis also showed that every element of the suggested framework makes its contribution to the overall performance. The use of subgroup discovery added to energy-based validation increased the diversity of synthetic samples and improved the quality of samples, respectively [12], [15]. Furthermore, the full ERMFM architecture significantly outperformed the various reduced versions of the architecture, emphasizing the value of combining Flow Matching, GMM clustering, EBM validation, and meta-learning in an integrated architecture [9], [11], [25].

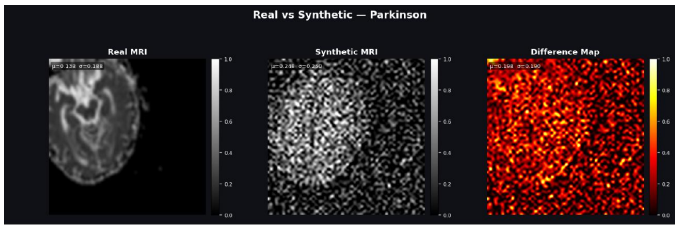


Fig. 10. Real vs Synthetic Parkinson

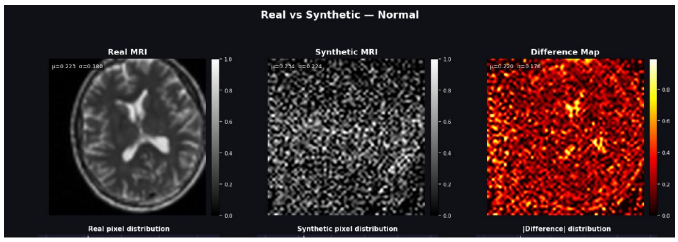


Fig. 11. Real vs Synthetic Normal

D. Discussion

The results of the experiments prove that the proposed ERMFM framework is able to solve the data scarcity problem in Parkinson's disease analysis [2], [5]. This approach of synthetic data generation using Flow Matching, subgroup clustering, validation using energy, and learning adaptation via meta-learning enhances classification accuracy and generalization capabilities [9], [11], [12], [15], [25]. The high ROC-AUC across few-shot settings suggests that the framework is robust even when encountering extreme label scarcity, promising its application in real-world healthcare settings [1], [2].

Various few-shot learning scenarios were tested to evaluate the proposed ERMFM framework. The experimental results present good classification accuracy of both the clinical and MRI datasets [1], [17]. The framework showed the highest ROC-AUC of 0.9490 and the highest accuracy of 0.9140, demonstrating good discrimination between patients with Parkinson's disease and healthy controls [34].

VI. CONCLUSION

This paper introduced an Energy-Regularized Meta-Learned Flow Matching (ERMFM) approach to tackle the data scarcity problem in Parkinson's Disease analysis. The proposed framework unites the generative modeling with the few-shot adaptation in one frame by adding the GMM-based subgroup discovery, the Flow Matching-based synthetic data generation, the Energy-Based Model validation and the iMAML-based few-shot adaptation. Experimental results on both clinical and MRI datasets showed good performances in terms of classification, up to a maximum ROC-AUC of 0.9490, while sustaining robust performances in various few-shot learning configurations. The results show that both using synthetic data generated using the group-aware method and using the energy-based validation method increases the diversity of the samples and improves the quality and consistency of the distribution of the samples. Moreover, iMAML allows effective adaptation with limited data, enhancing the model generalization

and robustness. Based on these results, it can be concluded that the proposed framework might be a viable solution to be adopted in the medical domain for creating trustworthy machine learning systems with limited data. The next step is to expand the framework to multimodal health data and assess its ability to other neurodegenerative diseases and prediction tasks in clinical practice.

VII. Future Work

Though the proposed Energy-Regularized Meta-Learned Flow Matching (ERMFM) framework demonstrated improvements in creation of synthetic data, adaptation to subgroups and the classification of PD, a few promising directions for future research are mentioned. First of all, the framework can be extended to include multimodal health care data, such as, but not limited to, genetic data, and data from wearable based sensors and EHSs and patient trajectories as they evolve over time. Combining different data modalities might enhance the model's ability to capture complex patterns in diseases and to gain a global perspective of a patient's heterogeneity.

Also, other subgroup discovery techniques, aside from GMMs, can be used to find additional subpopulations that are more clinically relevant, such as developing a deep clustering approach or a graph-based approach to learning a patient representation. There can also be a conditional/ personalized treatment generation mechanism to explicitly model responsiveness to treatment and disease progression, staging, and treatment response that can be implemented in the Flow Matching component. In addition, if the Energy-Based Model included the clinical knowledge of neurologists and clinical experts it would be more clinically relevant and practical.

The proposed framework will be expanded further to cover other neurological and chronic diseases to judge the applicability of the proposed framework to the other healthcare areas. In future, other learning techniques, however, can be applied which might be trainable in other medical institutions either individually or collectively, without giving the data such as federated learning or privacy-preserving learning. Lastly, clinical trials would need to be conducted in order to evaluate the utility, robustness and trustworthiness of the proposed framework for healthcare decision support systems.

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